

Transitivity of Positive Extreme-Point Hulls for Aligned Rearrangements

Solution packet for arXiv:1001.2426

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Status: likely valid substantial partial result. The general question in Remark 18 remains open. The new theorem proves transitivity whenever the intermediate rearrangement is obtained from the initial rearrangement by a nonincreasing multiplier in $[0, 1]$. A separate corollary records the full zero-dilation subcases already implicit in the source paper.

1 Source and exact question

The source is F. Sukochev and D. Zanin, *Orbits in symmetric spaces*, J. Funct. Anal. **257** (2009), no. 1, 194–218, arXiv:1001.2426 [1]. On page 12 of the compiled source PDF, immediately after Lemma 17, the authors ask:

The collection of sets $\{Q(x), x \in E\}$ also satisfies the transitivity property expressed in Lemma 17. We do not know whether this is the case for the collection $\{Q_+(x), x \in E\}$.

and this suffices to complete the proof. □

Remark 18. *The collection of sets $\{Q(x), x \in E\}$ also satisfies the transitivity property expressed in Lemma 17. We do not know whether this is the case for the collection $\{Q_+(x), x \in E\}$.*

4. MAIN RESULTS

Figure 1: Remark 18 in the source paper, rendered from the local source PDF.

Thus the question is whether

$$y \in Q_+(z), \quad z \in Q_+(x) \quad \implies \quad y \in Q_+(x)$$

for nonnegative elements of a symmetric Banach function space.

2 Definitions

Let $I = (0, 1)$ or $(0, \infty)$, let $E = E(I)$ be a fully symmetric Banach function space, and let f^* denote the nonincreasing rearrangement. For $x \geq 0$,

$$\Omega_+(x) = \{u \geq 0 : u \prec\prec x\}, \quad Q_+(x) = \overline{\text{conv}}^{\|\cdot\|_E} \text{extr } \Omega_+(x).$$

The source proves that on $(0, 1)$ the extreme rearrangements are $x^* \mathbf{1}_{[0, \beta]}$. On $(0, \infty)$ these remain all the extreme rearrangements when $x^*(\infty) = 0$; if $x^*(\infty) > 0$, there is also the full-rearrangement possibility described in Corollary 31 of the source.

We use the standard rearrangement-transfer fact on a nonatomic Lebesgue interval: a measure-preserving pullback preserves distributions, the norm of a symmetric space, and the extreme-point descriptions above. Consequently $Q_+(x)$ depends only on x^* and is invariant under equimeasurability. This is the same transfer principle used in the proof of Lemma 17 of the source.

3 Proof intuition

The useful feature of a nonincreasing multiplier $a \in [0, 1]$ is that its dyadic lower approximants are averages of indicators of initial intervals. Multiplying by x^* therefore turns each approximant into an average of extreme points of $\Omega_+(x)$. The approximation is uniform in the multiplier, so it converges in *every* symmetric norm; no order continuity is needed.

If $z^* = ax^*$, every extreme point of $\Omega_+(z)$ is obtained by one more initial-interval cut (or by the allowed full rearrangement). Its multiplier relative to x^* is still nonincreasing and lies in $[0, 1]$. Hence all generators of $Q_+(z)$ already lie in $Q_+(x)$, which proves the desired inclusion for this aligned class.

4 The aligned-multiplier theorem

Lemma 1 (Monotone multipliers lie in the positive hull). *Let $x \geq 0$ belong to E , and let $b : I \rightarrow [0, 1]$ be measurable and nonincreasing. If $w \geq 0$ satisfies*

$$w^* = bx^*,$$

then $w \in Q_+(x)$.

Proof. By rearrangement invariance it suffices first to treat $w = w^* = bx^*$. For $n \geq 1$ put

$$b_n = 2^{-n} \lfloor 2^n b \rfloor = 2^{-n} \sum_{j=1}^{2^n} \mathbf{1}_{\{b \geq j2^{-n}\}}.$$

Every level set of the nonincreasing function b is, up to a null set, an initial interval $[0, \beta_{n,j}]$. Therefore

$$b_n x^* = 2^{-n} \sum_{j=1}^{2^n} x^* \mathbf{1}_{[0, \beta_{n,j}]}$$

is a convex combination of extreme points of $\Omega_+(x)$, using the extreme-point characterization in the Appendix of [1]. The endpoints $\beta = 0$ and $\beta = \sup I$ cause no problem; in the semi-axis case x^* itself satisfies the source's full-rearrangement tail condition.

Since $0 \leq b - b_n < 2^{-n}$, the ideal property of E gives

$$\|(b - b_n)x^*\|_E \leq 2^{-n} \|x\|_E \longrightarrow 0.$$

Thus $bx^* \in Q_+(x)$. Finally, applying the standard measure-preserving rearrangement transfer to the approximating convex combinations gives the same conclusion for every w with $w^* = bx^*$. $\square$

Theorem 1 (Transitivity for aligned rearrangements). *Let $E = E(I)$ be fully symmetric, let $x, z \in E$ be nonnegative, and suppose that*

$$z^*(t) = a(t)x^*(t) \quad \text{a.e.}$$

for some nonincreasing measurable $a : I \rightarrow [0, 1]$. Then

$$z \in Q_+(x) \quad \text{and} \quad Q_+(z) \subseteq Q_+(x).$$

In particular, $y \in Q_+(z)$ implies $y \in Q_+(x)$.

Proof. The first assertion is Lemma 1 with $b = a$.

Let $v \in \text{extr } \Omega_+(z)$. On $(0, 1)$, and on $(0, \infty)$ for every truncated extreme point, the source's characterization gives

$$v^* = z^* \mathbf{1}_{[0, \gamma]} = (a \mathbf{1}_{[0, \gamma]}) x^*$$

for some γ . The multiplier $a \mathbf{1}_{[0, \gamma]}$ is again nonincreasing and takes values in $[0, 1]$, so Lemma 1 yields $v \in Q_+(x)$.

In the additional full-rearrangement case on $(0, \infty)$ one has $v^* = z^* = ax^*$, and the same lemma again gives $v \in Q_+(x)$. Thus every extreme point of $\Omega_+(z)$ belongs to the closed convex set $Q_+(x)$. Taking closed convex hulls proves $Q_+(z) \subseteq Q_+(x)$. $\square$

5 A full subcase already implicit in the source

Corollary 1 (Zero-dilation generators). *If $Q_+(x) = \Omega_+(x)$, then $Q_+(z) \subseteq Q_+(x)$ for every $z \in Q_+(x)$. In particular, the transitivity question has an affirmative answer whenever one of the following source hypotheses holds:*

- (i) $I = (0, 1)$ and $\varphi(x) = 0$;
- (ii) $I = (0, \infty)$, $E \subseteq L_1$, and $\varphi_{\text{fin}}(x) = 0$;
- (iii) $I = (0, \infty)$, $E \not\subseteq L_1$, and $\varphi(x) = 0$.

Proof. If $z \in Q_+(x) = \Omega_+(x)$ and $y \in Q_+(z) \subseteq \Omega_+(z)$, then $y \prec\prec z \prec\prec x$. Transitivity of submajorization gives $y \in \Omega_+(x) = Q_+(x)$. The three displayed hypotheses imply $Q_+(x) = \Omega_+(x)$ by Theorems 23, 25, and 27 of [1], respectively. $\square$

6 Verification, novelty, and limitations

Internal checks. The dyadic identity has exact coefficients summing to one. The only norm estimate is the lattice domination $|(b - b_n)x^*| \leq 2^{-n}x^*$, so the proof does not silently assume a Fatou norm, separability, or order continuity. The semi-axis proof separately checks the extra full-rearrangement extreme points.

Bounded novelty check. The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:1001.2426 and the Q_+ transitivity language. Exact-phrase and core-keyword searches of arXiv/web results, together with inspection of the closely related follow-up arXiv:1003.1817, found no explicit later resolution of Remark 18 and no statement of Theorem 1. This is a bounded search, so novelty confidence is moderate rather than definitive.

Limitation and exact obstruction. The theorem does not cover an arbitrary $z \in Q_+(x)$. Even $z \prec x$ need not imply $z^* \leq x^*$: mass may move from an early high plateau into a later region where x^* vanishes. In that situation z^*/x^* is not an admissible decreasing multiplier. Attempts to substitute convex approximations are blocked because initial-interval truncation is nonlinear and rearrangement of a sum need not preserve the aligned decomposition. This is precisely the remaining gap in the full problem.

Human-review recommendation. Check the rearrangement-transfer step in Lemma 1 against the preferred convention for measure-preserving transformations on the semi-axis, and verify that the full-tail extreme case is read exactly as in Corollary 31 of the source. The dyadic convex-hull mechanism itself is elementary and appears robust.

References

- [1] F. Sukochev and D. Zanin, *Orbits in symmetric spaces*, J. Funct. Anal. **257** (2009), no. 1, 194–218; [arXiv:1001.2426](#).
- [2] S. G. Krein, Yu. I. Petunin, and E. M. Semenov, *Interpolation of Linear Operators*, Translations of Mathematical Monographs 54, American Mathematical Society, 1982.